We now return to the question of how to find the 'volume under the surface.' If the region we are defined on, R is *nicer* to view via polar coordinates, then we want to do so – but how do we then compute the resulting integral? We return to the beginning and discuss the notion of *simplicity*:

Generally speaking, most polar regions are given as r-simple regions. The idea of 'upper' and 'lower' bounds still holds, but now we are talking about them in relation to either the pole (r-simple) or the horizontal axis (θ-simple).

Now, recall how we converted from Cartesian to polar. In part, we used the following equations:

$$x = r\cos(\theta) \qquad \text{and} \qquad y = r\sin(\theta).$$

So, given a Cartesian surface $f(x, y)$, we can rewrite it as $f(r, \theta) = f(r\cos(\theta), r\sin(\theta))$.

Now, if we want to compute the 'volume under the surface' of $f(x, y)$ over R, we can do the following:

Theorem. [Fubini's Theorem (Polar Form)] Let $f(x, y)$ be a continuous surface defined on a region R with polar representation $f(r, \theta)$.

(i) If R is r-simple, then $\displaystyle\iint_R f(x, y)\, dA = \int_a^b \int_{g_1(\theta)}^{g_2(\theta)} f(r, \theta)\, r\, dr\, d\theta.$

(ii) If R is θ-simple, then $\displaystyle\iint_R f(x, y)\, dA = \int_a^b \int_{h_1(r)}^{h_2(r)} f(r, \theta)\, r\, d\theta\, dr.$

NOTE: Observe that the polar integral has an extra 'r' in its differential. This is incredibly important and is easy to forget!

Example. Find the volume under the surface $f(x, y) = e^{x^2+y^2}$ over the region R, where R is the semicircular region bounded by the x-axis and the curve $y = \sqrt{1 - x^2}$.

Example. Compute the following integral: $\displaystyle\int_0^1 \int_0^{\sqrt{1-x^2}} x^2 + y^2 \, dy \, dx$.

Example. Compute the following integral: $\displaystyle\int_0^7 \int_0^y x \, dx \, dy$.

Just as we did before, we can use this volume idea to compute 2-dimensional area by simply integrating the surface $f(x, y) = 1$. Note that its polar representation is still just $f(r, \theta) = 1$, so

$$\text{Area of } R = \iint_R r \, dr \, d\theta \, .$$

Example. Find the area of a single petal on the three-petaled rose $r = \cos(3\theta)$.

Example. Find the area enclosed by the lemniscate $r^2 = 4\cos(2\theta)$.